# VEDANT SUKHADIA

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## SUMMARY

Machine Learning Engineer with over 3 years of hands-on experience building and deploying end-to-end ML solutions that solve real-world problems. I've developed real-time recommendation engines, forecasting models, and NLP applications using tools like TensorFlow, PyTorch, Scikit-learn, and Huggingface. I enjoy working with deep learning architecture (CNNs, RNNs, Transformers) and creating scalable data pipelines with Spark, Airflow, Hive, and BigQuery. I'm comfortable deploying models in Docker/Kubernetes environments via MLflow, SageMaker, and REST APIs. I thrive in Agile/Scrum teams, translating business needs into smart, data-driven solutions. I also focus on MLOps, with strengths in model monitoring, A/B testing, and explainability using SHAP and TensorBoard.

## TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Java, Scala
- **ML & Deep Learning Frameworks:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Huggingface
- **Modeling & Algorithms:** Deep Learning (CNNs, RNNs, Transformers), R Systems, Time Series Forecasting, NLP.
- **Data Processing:** Pandas, NumPy, Spark, Airflow
- **MLOps & Cloud Platforms :** MLflow, SageMaker, Docker, Kubernetes, REST APIs, AWS, GCP
- **Experimentation & Monitoring:** A/B Testing, SHAP, TensorBoard
- **Version Control & Workflow:** Git, Agile/Scrum

## WORK EXPERIENCE

**Machine Learning Engineer Intern @** CVS Health - USA **Dec 2024 – May 2025**

- Built a real-time product recommendation engine using PyTorch & XGBoost, boosting OTC conversions by 12% & CTR by 18%.
- Developed scalable data pipelines with Spark and Airflow, reducing processing latency by 35%.
- Deployed ML workflows via MLflow and AWS SageMaker within Docker/Kubernetes environments.
- Integrated SHAP-based model interpretability to ensure clinical compliance and transparency.
- Collaborated in Agile teams with data science, DevOps, and marketing to iteratively improve models.

**Machine Learning Engineer @** Infosys - India **Jun 2020 – July 2023**

- Developed demand forecasting models (RNN, regression) improving forecast accuracy by 15% and reducing stockouts by 12%.
- Engineered features from sales, inventory, and external data using Pandas, NumPy, and Hive.
- Integrated ML outputs into ERP systems using REST APIs for automated planning.
- Containerized training environments with Docker, streamlining deployment workflows.
- Automated validation/reporting, reducing QA effort by 40%; actively contributed to Agile ceremonies.

## PROJECT WORK

### Smart Health FAQ Bot using BERT and Flask

- Fine-tuned a pretrained BERT model using a domain-specific healthcare Q&A dataset to enhance semantic understanding and ensure contextually accurate responses for patient queries.
- Designed and deployed the chatbot as a RESTful web application using Flask, integrating MongoDB to manage conversation history, user feedback, and intent-logging for future model improvements.

### Image Classification with Transfer Learning (ResNet-50)

- Implemented an image classification pipeline using transfer learning on the ResNet-50 architecture to categorize plant leaf images into various disease classes based on visual symptoms.
- Applied advanced image preprocessing, data augmentation techniques, and model fine-tuning using TensorFlow and OpenCV, running training across multiple GPU environments to accelerate performance.

## EDUCATION

**University of New Haven** | Master's in Information Science **| GPA: 3.6/4.0** West Haven, Connecticut
Courses: Data analytics, database systems, machine learning, cloud computing, data visualization, and IT project management. 2023 – 2025

**Nirma University** | B. Tech in Electronics and Communication **| GPA: 7.3/10** Ahmedabad, Gujarat
Courses: Signal processing, embedded systems, communication theory, microcontrollers, circuit design, VLSI, RF engineering, and IoT systems. 2018 – 2022

## SELECTED PUBLICATIONS

- Sukhadia, V. K., & Sharma, R. (2024). A Scalable Framework for Real-Time Product Recommendations Using Deep Learning. **Proceedings of the International Conference on Applied Machine Learning**, 34(2), 112–120.
- Sukhadia, V. K. (2023). Interpretable AI in Healthcare: Integrating SHAP Values in Clinical Decision Systems. **Journal of Machine Learning and Data Science**.
- Sukhadia, V. K., & Smith, A. (2024). Optimizing Real-Time Recommendation Systems Using Transformer Architectures and Reinforcement Learning. **Proceedings of the International Conference on Machine Learning Systems (ICMLS)**, New York.

## CERTIFICATIONS

- **Machine Learning Specialization** – DeepLearning.AI
- **Deep Learning Specialization** – DeepLearning.AI
- **Data Science Toolbox** – Johns Hopkins University (Coursera)